

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{12} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[\mathbf{m}_d^{\mathbf{P}}] - \frac{1}{12} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_d^{\mathbf{P}}] - \frac{1}{2} g_1^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,0}[\mathbf{m}_d^{\mathbf{r}}] + \frac{1}{2} g_1^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \right. \\ & \quad \text{LF}_{4,-1}[\mathbf{m}_d^{\mathbf{r}}] + \frac{1}{4} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[\mathbf{m}_e^{\mathbf{P}}] - \frac{1}{4} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_e^{\mathbf{P}}] - \frac{1}{2} g_1^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,0}[\mathbf{m}_e^{\mathbf{r}}] + \\ & \quad \frac{1}{2} g_1^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{4,-1}[\mathbf{m}_e^{\mathbf{r}}] + \frac{1}{8} (g_1^2 + g_2^2) (2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_{\mathbf{p}} (g_1^2 - g_2^2)) \text{LF}_{3,0}[\mathbf{m}_l^{\mathbf{P}}] - \\ & \quad \frac{1}{8} (g_1^2 + g_2^2) (2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_{\mathbf{p}} (g_1^2 - g_2^2)) \text{LF}_{4,-1}[\mathbf{m}_l^{\mathbf{P}}] + \\ & \quad \frac{1}{24} (-6 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} (g_1^2 - 3 g_2^2) + 6 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + 3 g_2^2) + \sum_{\mathbf{p}} (g_1^4 - 9 g_2^4)) \text{LF}_{3,0}[\mathbf{m}_q^{\mathbf{P}}] + \\ & \quad \frac{1}{24} (6 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} (g_1^2 - 3 g_2^2) - 6 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} (g_1^2 + 3 g_2^2) - \sum_{\mathbf{p}} (g_1^4 - 9 g_2^4)) \text{LF}_{4,-1}[\mathbf{m}_q^{\mathbf{P}}] + \\ & \quad \frac{1}{3} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[\mathbf{m}_u^{\mathbf{P}}] - \frac{1}{3} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_u^{\mathbf{P}}] - g_1^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,0}[\mathbf{m}_u^{\mathbf{r}}] + g_1^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{4,-1}[\mathbf{m}_u^{\mathbf{r}}] - \\ & \quad \frac{7}{16} g_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{8} g_1^4 m_1^2 \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] + g_1^4 \text{LF}_{3,2,-2}[\mathbf{m}_1, \tilde{\mu}] + \\ & \quad \frac{3}{4} g_1^4 m_1^2 \text{LF}_{3,2,-1}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{2} g_1^4 \text{LF}_{3,3,-3}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{2} g_1^4 m_1^2 \text{LF}_{3,3,-2}[\mathbf{m}_1, \tilde{\mu}] - \\ & \quad \frac{1}{2} g_1^4 \text{LF}_{4,2,-3}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{2} g_1^4 m_1^2 \text{LF}_{4,2,-2}[\mathbf{m}_1, \tilde{\mu}] - \frac{35}{16} g_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] - \\ & \quad \frac{1}{8} g_2^4 m_2^2 \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + 5 g_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] + \frac{3}{4} g_2^4 m_2^2 \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] - \\ & \quad \frac{5}{2} g_2^4 \text{LF}_{3,3,-3}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{2} g_2^4 m_2^2 \text{LF}_{3,3,-2}[\mathbf{m}_2, \tilde{\mu}] - \frac{5}{2} g_2^4 \text{LF}_{4,2,-3}[\mathbf{m}_2, \tilde{\mu}] - \\ & \quad \frac{1}{2} g_2^4 m_2^2 \text{LF}_{4,2,-2}[\mathbf{m}_2, \tilde{\mu}] + \frac{1}{2} g_1^2 (-\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_d^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{P}}] + \\ & \quad \frac{1}{4} g_1^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} - \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_d^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{P}}] + \frac{1}{2} g_1^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} - \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \\ & \quad \text{LF}_{4,1,-1}[\mathbf{m}_d^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{P}}] + \frac{1}{2} g_1^2 (-\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_e^{\mathbf{r}}, \mathbf{m}_l^{\mathbf{P}}] + \\ & \quad \frac{1}{4} g_1^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} - \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_e^{\mathbf{r}}, \mathbf{m}_l^{\mathbf{P}}] + \frac{1}{2} g_1^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} - \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \\ & \quad \text{LF}_{4,1,-1}[\mathbf{m}_e^{\mathbf{r}}, \mathbf{m}_l^{\mathbf{P}}] + \frac{1}{8} (g_1^2 + g_2^2) (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} - \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_l^{\mathbf{P}}, \mathbf{m}_e^{\mathbf{r}}] - \\ & \quad \frac{1}{8} (g_1^2 - 3 g_2^2) (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} - \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\mathbf{P}}, \mathbf{m}_d^{\mathbf{r}}] - \\ & \quad \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_q^{\mathbf{P}}, \mathbf{m}_q^{\mathbf{s}}] + \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\mathbf{P}}, \mathbf{m}_q^{\mathbf{s}}] + \\ & \quad \frac{1}{4} (g_1^2 + 3 g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_q^{\mathbf{P}}, \mathbf{m}_u^{\mathbf{r}}] - \\ & \quad \frac{1}{8} (g_1^2 + 3 g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\mathbf{P}}, \mathbf{m}_u^{\mathbf{r}}] - \\ & \quad \frac{1}{4} (g_1^2 + 3 g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\mathbf{P}}, \mathbf{m}_u^{\mathbf{r}}] - \\ & \quad \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_q^{\mathbf{s}}, \mathbf{m}_q^{\mathbf{P}}] + \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\mathbf{s}}, \mathbf{m}_q^{\mathbf{P}}] + \\ & \quad \frac{1}{2} g_1^2 (-\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{P}}] + \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_u^{\mathbf{t}}, \mathbf{m}_d^{\mathbf{r}}] - \\ & \quad \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_u^{\mathbf{t}}, \mathbf{m}_d^{\mathbf{r}}] + \frac{3}{8} g_1^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \\ & \quad \frac{1}{8} g_1^4 m_1^2 \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{8} g_1^4 \text{LF}_{4,2,-3}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{8} g_1^4 m_1^2 \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \\ & \quad \frac{15}{8} g_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \frac{1}{8} g_2^4 m_2^2 \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_2] - \frac{5}{8} g_2^4 \text{LF}_{4,2,-3}[\tilde{\mu}, \mathbf{m}_2] - \\ & \quad \frac{1}{8} g_2^4 (m_2^2 - 12 \tilde{\mu}^2) \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_2] - 2 g_1^2 g_2^2 \text{LF}_{2,2,1,-2}[\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_1] - \\ & \quad \frac{3}{2} g$$